

Algebra 1: Unit 11, Lesson 4

1. A gardener plans on doubling his sunflower population each year. Currently, he has 8 sunflower plants. The exponential function that represents this growth is $S(t) = 8(2)^t$, where t is the time, in years, and $S(t)$ is the amount of sunflower plants.

What is the domain for $S(t)$?

2. Caffeine takes time to leave the system. Esther drank a cup of coffee this morning. The function that represents the amount of caffeine remaining in her body after drinking her coffee is given by $g(t) = 95(0.889)^t$ where t is the time in hours and $g(t)$ is the amount of caffeine remaining in milligrams (mg).

What is the interpretation of the y -intercept $(0,95)$ in this context?

3. A new copy machine that was recently purchased depreciates over time by the given function, $C(t) = \$5,500\left(\frac{1}{6}\right)^t$ where t is the time in years and $C(t)$ is the value of the copy machine.

What is the end behavior of $C(t)$?